

VILÉM ZOUHAR

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EDUCATION

ETH zürich

ETH Zürich

2022-Dec 2026 (expected)

PhD in Computer Science in **LRE lab**

Advisors: Mrinmaya Sachan, Menna El-Assady

Thesis: Prudent & Bespoke AI Evaluation



UNIVERSITÄT
DES
SAARLANDES



university of
 groningen

Saarland University + Groningen University

Sep 2020 - Jun 2022

MSc. in Language Science and Technology (double)

Advisors: Dietrich Klakow, Gosse Bouma

Thesis **Shrinking Knowledge Base Size: Dimension Reduction & Filtering**



Charles University in Prague

Sep 2017 - Jun 2020

BSc. in Computer Science (computational linguistics minor)

Advisor: Ondřej Bojar

Thesis **Enabling Outbound Machine Translation**



RESEARCH INTERESTS (inter alia)

▲ Formal methods of AI evaluation

„Without evaluation, science is limited to blind exploration.“

▲ Practical AI evaluation also with humans

„How to make humans judge quality?“

▲ Efficient AI evaluation

How to compare models robustly yet economically?



WORK EXPERIENCE

▲ **Google Translate**, Mountain View (research intern, evaluation)

Jun-Oct 2025

▲ Microsoft Translate, remote (applied scientist intern, model evaluation)

Jun-Oct 2024

▲ **Amazon Machine Translation**, New York (applied scientist intern, quality estimation)

Jun-Oct 2023

▲ **Spoken Language Systems**, Saarland University (student research assistant)

Jan-Dec 2021

▲ **Institute of Formal and Applied Linguistics**, Charles University (student research assist.)

Oct 2019 - Jun 2022

▲ Previo (software dev intern)

Jun-Sep 2018

▲ BIM Project (software dev intern)

Jun-Sep 2017



AWARDS AND SCHOLARSHIPS

▲ Cohere Labs Catalyst grant for efficient evaluation

2026

▲ Google PhD Fellowship in Natural Language Processing

2025

▲ Two EAMT grants for research on compute-efficient evaluation, and LLM brittleness

2025, 2025

▲ Language & Communication Technologies full MSc Erasmus Mundus **scholarship**

2020-2022

▲ Student faculty grant for **learning materials** for Machine Learning Exercises in R

2018



TEACHING

▲ Human evaluation tutorial

EAMT 2026, KONVENS 2026, MT Marathon 2026

▲ Large Language Models TA

2023, 2024, 2025, 2026 (+lecture)

▲ Deep Learning TA

2025

▲ Interactive Machine Learning Visualization and Explainability TA

2025

▲ Neural Networks Implementation and Applications head TA

2021

▲ Statistical Natural Language Processing head TA

2021

Contrastive ESA: Human Evaluation of Multiple Translations at Once (WMT 2026)

Vilém Zouhar, Roman Grundkiewicz, Sara Rajae, Parker Riley, Martin Popel, Rachel Bawden, Philipp Koehn, Marine Carpuat, Tom Kocmi

Pearmut: Human Evaluation of Translation Made Trivial (EMNLP 2026 Demo)

Vilém Zouhar, Tom Kocmi

How to Select Datapoints for Efficient Human Evaluation of NLG Models? (TACL 2025)

Vilém Zouhar, Peng Cui, Mrinmaya Sachan

Early-Exit and Instant Confidence Translation Quality Estimation (EACL 2026)

Vilém Zouhar, Maike Züfle, Beni Egressy, Julius Cheng, Jan Niehues

Generating Difficult-to-Translate Texts (MME at EACL 2026)

Vilém Zouhar, Wenda Xu, Parker Riley, Juraj Juraska, Mara Finkelstein, Markus Freitag, Dan Deutsch

AI-Assisted Human Evaluation of Machine Translation (NAACL 2025)

Vilém Zouhar,[✉] Tom Kocmi,[✉] Mrinmaya Sachan

Error Span Annotation: A Balanced Approach for Human Evaluation of Machine Translation (WMT 2024)

Tom Kocmi,[✉] **Vilém Zouhar**,[✉] Eleftherios Avramidis, Roman Grundkiewicz, Marzena Karpinska, Maja Popović, Mrinmaya Sachan, Mariya Shmatova

Quality and Quantity of Machine Translation References for Automated Metrics (HumEval 2024)

Vilém Zouhar, Ondřej Bojar

Navigating the Metrics Maze: Reconciling Score Magnitudes and Accuracies (ACL 2024)

Tom Kocmi, **Vilém Zouhar**, Christian Federmann, Matt Post

Distributional Properties of Subword Regularization (EMNLP 2024)

Marco Cognetta, **Vilém Zouhar**, Naoaki Okazaki

A Diachronic Perspective on User Trust in AI under Uncertainty (EMNLP 2023)

Shehzaad Dhuliawala,[✉] **Vilém Zouhar**,[✉] Mennatallah El-Assady, Mrinmaya Sachan

Tokenization and the Noiseless Channel (ACL 2023)

Vilém Zouhar, Clara Meister, Juan Luis Gastaldi, Li Du, Mrinmaya Sachan, Ryan Cotterell

Re-visiting Automated Topic Model Evaluation with Large Language Models (EMNLP 2023)

Dominik Stambach, **Vilém Zouhar**, Alexander Hoyle, Mrinmaya Sachan, Elliott Ash

Neural Machine Translation Quality and Post-Editing Performance (EMNLP 2021)

Vilém Zouhar, Ondřej Bojar, Martin Popel, Aleš Tamchyna

Dynamically Allocating Evaluation Effort for Model Ranking (EMNLP 2026)

Vilém Zouhar, Julia Kreutzer, Alon Lavie, Tom Kocmi, Matt Post, Ondřej Bojar, Mrinmaya Sachan

Last Translation Benchmark (in progress 2026)

Vilém Zouhar and >200 co-authors

Estimating Machine Translation Difficulty (EMNLP 2025)

Lorenzo Proietti,[✉] Stefano Perrella,[✉] **Vilém Zouhar**,[✉] Roberto Navigli, Tom Kocmi

Findings of the WMT25 general machine translation shared task: Time to stop evaluating on easy test sets (WMT 2025)

Tom Kocmi, **V. Zouhar** and others

Searching for Difficult-to-Translate Test Examples at Scale (In review 2026)

Wenda Xu,[✉] **Vilém Zouhar**,[✉] Parker Riley, Mara Finkelstein, Markus Freitag, Daniel Deutsch

Pitfalls and Outlooks in Using COMET (WMT 2024)

Vilém Zouhar,[✉] Pinzhen Chen,[✉] Tsz Kin Lam, Nikita Moghe, Barry Haddow

Fine-Tuned Machine Translation Metrics Struggle in Unseen Domains (ACL 2024)

Vilém Zouhar, Shuoyang Ding, Anna Currey, Tatyana Badeka, Jenyuan Wang, Brian Thompson

WMT24 General Machine Translation Shared Task: The LLM Era is Here but MT is Not Solved Yet (WMT 2024)

Tom Kocmi, **V. Zouhar** and others

RELIC: Investigating Large Language Model Responses using Self-Consistency (CHI 2024)

Furui Cheng, **Vilém Zouhar**, Simran Arora, Mrinmaya Sachan, Hendrik Strobelt, Mennatallah El-Assady

Evaluating Optimal Reference Translations (JNLE 2024)

Vilém Zouhar, Věra Kloudová, Martin Popel, Ondřej Bojar

WMT 2023 Shared Task on Machine Translation with Terminologies (EMNLP 2023)

Kirill Semenov, **Vilém Zouhar**, Tom Kocmi, Dongdong Zhang, Wangchunshu Zhou, Yuchen Eleanor Jiang

A Formal Perspective on Byte-Pair Encoding (ACL 2023)

Vilém Zouhar, Clara Meister, Juan Luis Gastaldi, Li Du, Tim Vieira, Mrinmaya Sachan, Ryan Cotterell

Poor Man's Quality Estimation: Predicting Ref.-Based MT Metrics Without Reference (EACL 2023)

Vilém Zouhar, Shehzaad Dhuliawala, Wangchunshu Zhou, Nico Daheim, Tom Kocmi, Yuchen Eleanor Jiang, Mrinmaya Sachan

Providing Backtranslation Improves Users Confidence in MT, Not Quality (NAACL 2021)

V. Zouhar, M. Novák, M. Žilínek, O. Bojar, M. Obregón, R. L. Hill, F. Blain, M. Fomicheva, L. Specia, L. Yankovskaya

Generating Adversarial Texts for Machine Translation via GRPO (KONVENS 2026)

Why We Need Speech to Evaluate Speech Translation (EMNLP 2026)

When AI Benchmarks Plateau: A Systematic Study of Benchmark Saturation (ICML 2026)

How Important is 'Perfect' English for Machine Translation Prompts? (EACL 2026)

Hearing to Translate: The Effectiveness of Speech Modality Integration into LLMs (TACL 2026)

Findings of the WMT25 shared task on automated translation evaluation systems: Linguistic diversity is challenging and references still help (WMT 2025)

COMET-poly: Machine Translation Metric Grounded in Other Candidates (WMT 2025)

Co-DETECT: Collaborative Discovery of Edge Cases in Text Classification (EMNLP 2025 Demo)

Biased Tales: Cultural and Topic Bias in Generating Children's Stories (EMNLP 2025)

QE4PE: Word-level Quality Estimation for Human Post-Editing (TACL 2025)

Hurdles of Automatic Metric for Speech Translation Evaluation (IWSLT 2026)

Findings of the IWSLT 2025 Evaluation Campaign (IWSLT 2025)

Understanding Large Language Model Behaviors Through Interactive Counterfactual Generation and Analysis (IEEEvis 2025)

Two Counterexamples to Tokenization and the Noiseless Channel (LREC-COLING 2024)

Sampling and Filtering of Neural Machine Translation Distillation Data (NAACL SRW 2021)

Harmonizing Assistance: Moderating Visual and Textual Aids in AI-Enhanced Textbook Reading with IRead (IJAIED 2025)

AutoTutor meets Large Language Models: A Language Model Tutor with Rich Pedagogy and Guardrails (Learning@Scale 2024)

Artefact Retrieval: Overview of NLP Models with Knowledge Base Access (AKBC CSKB 2021)

Sentence Ambiguity, Grammaticality and Complexity Probes (BlackboxNLP 2022)

There is No Theoretical Curse of Multilinguality For Embedding Space Structure (in review 2026)

Augmenting Text to Increase Translation Difficulty (EAMT 2026)

The Role of Disfluencies in Speech Translation (in review 2026)

Can Large Language Models Capture Human Annotator Disagreements? (EACL 2026)

Findings of the WMT25 Multilingual Instruction Shared Task: Persistent Hurdles in Reasoning, Generation, and Evaluation (WMT 2025)

Findings of the WMT25 Terminology Translation Task: Terminology is Useful Especially for Good MTs (WMT 2025)

Deconstructing Self-Bias in LLM-generated Translation Benchmarks (EMNLP 2026)

Unsupervised Word-level Quality Estimation for Machine Translation Through the Lens of Annotators (Dis)agreement (EMNLP 2025)

Large Language Models as Span Annotators (MME at EACL 2026)

A Bayesian Optimization Approach to Machine Translation Reranking (NAACL 2025)

Speech Translation and Metrics in 2026: Findings of the IWSLT Campaign (IWSLT 2026)

Are Large Language Models for Education Reliable for All Languages? (BEA 2025)

PWESuite: Phonetic Word Embeddings and Tasks They Facilitate (LREC-COLING 2024)

Knowledge Base Index Compression via Dimensionality and Precision Reduction (SpaNLP 2022)

WMT20 Document-Level Markable Error Exploration (WMT 2020)

How to Engage Your Readers? Generating Guiding Questions to Promote Active Reading (ACL 2024)

Enhancing Textbooks with Visuals from the Web for Improved Learning (EMNLP 2023)

Leveraging Neural Machine Translation for Word Alignment (PBML 116)

INVITED TALKS

- ▲ Theory and Practice of Efficient Evaluation at Google Gboard (2026), SICSA (2026)
- ▲ Selecting Examples to Human-Evaluate NLG Models at Charles University (2025), TU Darmstadt (2025), UMD Multilingual Seminar (2026)
- ▲ Prudent NLG Evaluation at KIT (2025), Cardiff NLP Seminars (2025), Google Translate (2024)
- ▲ Token(s) of Appreciation for BPE at MT Marathon (2024)
- ▲ ESA and ESAAI at Microsoft Translate (2024)
- ▲ How we solved tokenization but got it wrong at ZurichNLP Meetup #9 (2024)
- ▲ Quality and Quantity of Machine Translation References for Automated Metrics at IST/Unbabel seminar (2024)
- ▲ Poor Man's Quality Estimation at IST/Unbabel seminar (2023)

SERVICE

- ▲ **General Machine Translation** Shared Task at WMT (co-organizer) 2024, 2025, 2026
- ▲ **Evaluation Metrics** Shared Task at WMT (co-organizer) 2025, 2026
- ▲ **Multilingual Instruction** Shared Task at WMT (co-organizer) 2025, 2026
- ▲ **Multilingual Multicultural Evaluation Workshop** at EACL (organizer) 2026
- ▲ **SIGTYP Workshop: Computational Linguistic Typology and Multilingual NLP** at EACL (co-organizer) 2026
- ▲ **ACL Mentorship Panel**: Navigating Mental Health in Fast-Paced AI Research (panelist) 2026
- ▲ **WMT Terminology** Shared Task (organizer) 2023, 2025
- ▲ Evaluation committee member for granting university accreditations in Czechia 2020-2025
- ▲ **CSRR**: Workshop on Commonsense Representation and Reasoning (organizer) ACL 2022
- ▲ Area chair GEM 2026, EMNLP Demo 2026, ARR 2026, EAMT 2027
- ▲ Reviewer: {EACL, MME, ACL, ARR, SwissText} 2026
{NAACL★, ACL, EMNLP, WMT, ARR} 2025 {LREC-COLING, EACL★, ARR, Repl4NLP, EMNLP, WMT} 2024
{ARR, WMT, EMNLP, AACL-IJCNLP, ACL} 2023 {SVRHM, CoNLL, AACL-IJCNLP} 2022

TECHNICAL KNOWLEDGE

[GitHub](#)

LANGUAGES

- ▲ **Programming**: Python, JS/TS, Rust
- ▲ **Toolkits**: LM eval harness, vllm, PyTorch, Scikit, Numpy, HF, Fairseq, Marian
- ▲ **Misc**: Linux (long-term user, cluster), passionate about visualization, writing and typesetting (Typst, LaTeX)
- ▲ Czech: C2
- ▲ English: C2
- ▲ German: B2
+ linguistic curiosity

STUDENTS

Pleasure to (co-)supervise student projects/theses

- ▲ Rosamund Ang: Sample-Efficient Aligning of Automatic Metrics ETH 2026
- ▲ Šimon Sukup, William Kalikman, Michal Tešnar: Augmenting Texts to Increase Translation Difficulty ETH 2026
- ▲ Victor Zarzu: Speech Translation Quality Estimation ETH 2026
- ▲ Chenfei Xiong: Collaborative Discovery of Edge Cases in Text Classification ETH 2025
- ▲ Oymak Gül, Hatipoglu Öykü, Garries Urbina Oscar, Victor Zarzu Interactive Visualization of Chain-of-Thought Reasoning in Large Language Models ETH 2025
- ▲ Yijie Tong, Haokun He (Self Quality Estimation for Machine Translation and NLP) UZH 2024
- ▲ Abhinav Kumar (Data Augmentation for Text Complexity Prediction) UZH 2024
- ▲ Vincent Dörig (IRead: AI-Enhanced Textbook Reading) ETH 2024
- ▲ David Gu (Manifestations of Image Complexity) ETH 2023

EXTRA-CURRICULAR

- ▲ Academic senate at Charles University Faculty of Mathematics and Physics 2018-2020
- ▲ Several **games** programmed and presented in limited time, mostly Ludum Dare 2015-present
- ▲ Organization of **Kasiopea**, an annual coding competition for talented high school students 2017-2019
- ▲ Amateur electric guitar  2021-present